



IMSE 785

Big Data Analytics

Shing Chang

IMSE, K-State

Lecture 1 Intro to Big Data Analytics

- What is Big Data
- What is Big Data Analytics
- Why does Big Data Analytics matter
- Learning Big Data Analytics
- Big Data Analytics Examples
 - Process Big Data Analytics
 - Business Big Data Analytics

What is Big Data

- **Big data** is a collection of data sets **so large** and **complex** that it becomes difficult to process using on-hand database management tools or traditional data processing applications. (source: http://en.wikipedia.org/wiki/Big_data)
- IDC estimate put the size of the “digital universe” at 0.18 zettabytes in 2006 and is forecasting a tenfold growth by 2011 to 1.8 zettabytes. (source: Hadoop: the definitive guide, 3rd edition)
- ***IDC now forecasts that we will generate 40 zettabytes (ZB) by 2020.***
- Five exabytes of information was generated from the dawn of time to 2003 (Source: Intel)
- In 2012 along, the global data grows to 2.7 zettabytes. (Source: Intel)
- A zettabyte is 10^{21} bytes, or equivalently one thousand exabytes, one million petabytes, or one billion terabytes.

Definition of various byte

<< A gigabyte is equal to 1,024 megabytes.

A terabyte is equal to 1,024 gigabytes.

A petabyte is equal to 1,024 terabytes.

An exabyte is equal to 1,024 petabytes.

A zettabyte is equal to 1,024 exabytes.

A yottabyte is equal to 1,024 zettabytes.

*From there, we'll be working backwards through the Latin alphabet
(there simply weren't enough Titans in Greek mythology to go around),
until we reach the alphabyte.*

*What ever comes after that isn't worth considering – I
t's just a helluva-lottabytes! >>*

****** quoted from a pc encyclopedia

USER STORY THERE ARE VARIOUS WAYS TO QUANTIFY BIG DATA.

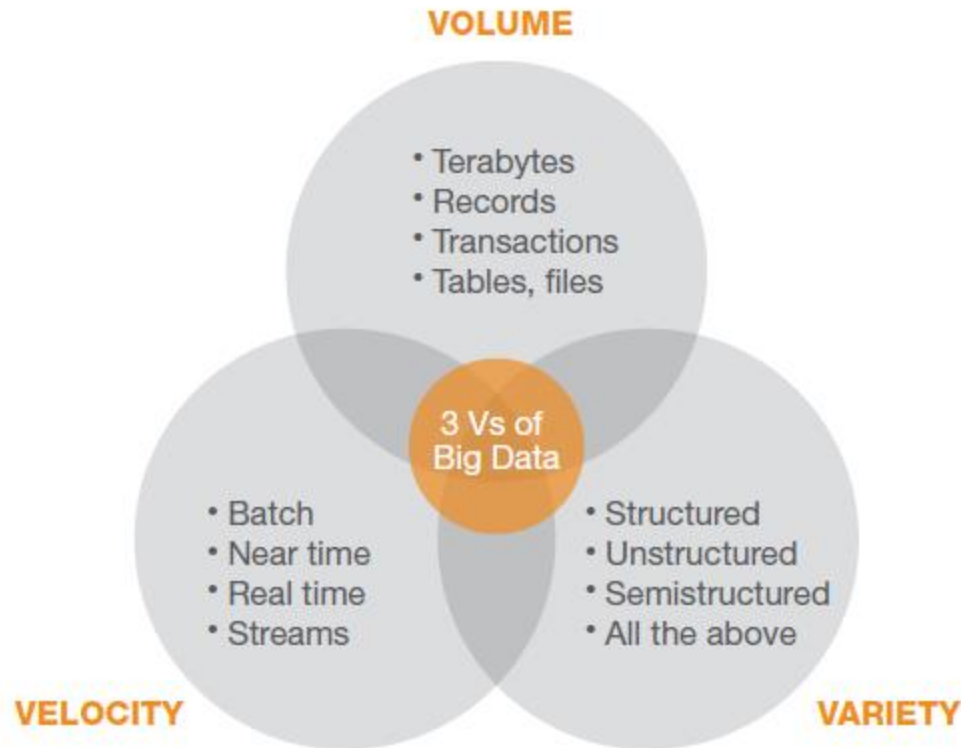
TDWI asked a user how many terabytes he's managing for analytics, and he said: "I don't know, because I don't have to worry about storage. IT provides it generously, and I tap it like crazy." Another user said: "We don't count terabytes. We count records. My analytic database for quality assurance alone has 3 billion records. There's another 3 billion in other analytic databases."

2011 TDWI report

Other Big Data Definitions

- Rather than using how many bytes some define Big Data using how many variables and records
- If you cannot analyze the data you collected in time using traditional processes or tools, it is BIG DATA for you!
 - E.g. two hundred variables with 100,000 records each
 - E.g. two thousand variables with tens of millions of records each month for Medicare claims

Defining Big Data via the Three Vs



Source: Table 1. 2011 TDWI report sponsored by SAS

Examples of Big Data

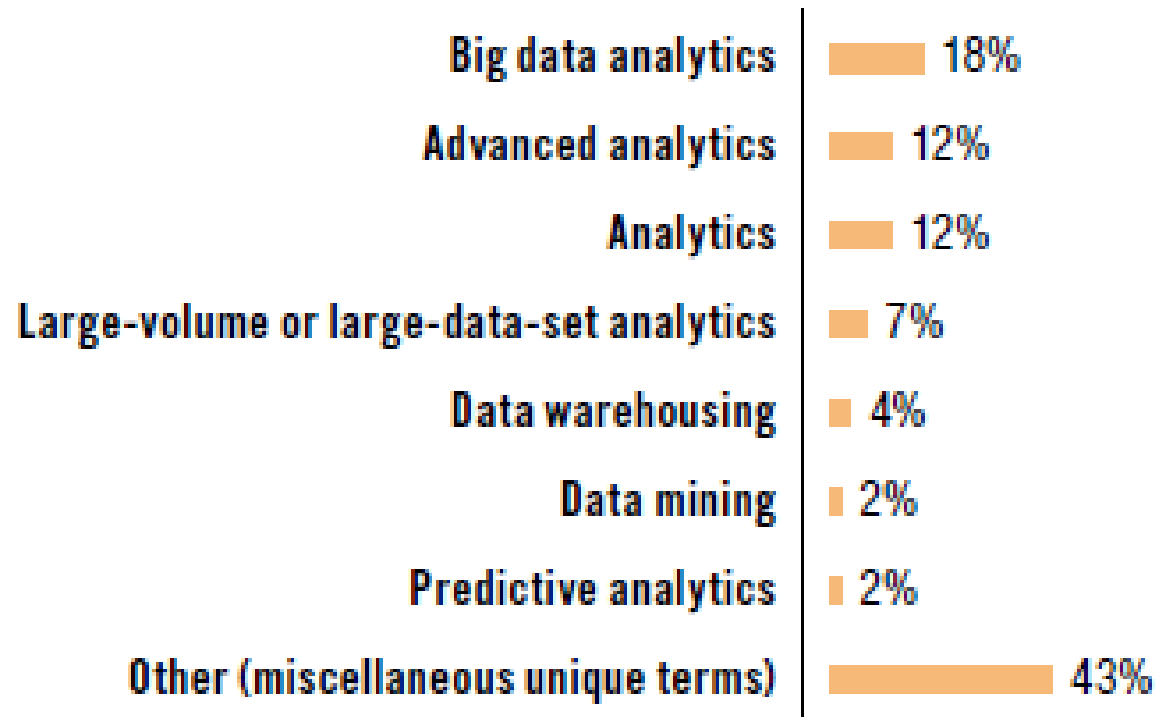
- The New York Stock Exchange generates about one terabyte of new trade data per day.
- Facebook hosts approximately 10 billion photos, taking up one petabyte of storage.
- Ancestry.com, the genealogy site, stores around 2.5 petabytes of data.
- The Internet Archive stores around 2 petabytes of data and is growing at a rate of 20 terabytes per month.
- The Large Hadron Collider near Geneva, Switzerland, will produce about 15 petabytes of data per year.

(source: Hadoop: the definitive guide, 3rd edition)

Big Data Analytics

- Big data analytics is where advanced analytic techniques operate on big data
- Consider the following analysis problems:
 - When you try to load a big file into Excel, a system error happens
 - When you try to multiply a matrix to a vector, your computer CPU runs out of memory
 - It will take at least one week to just download all data from a public domain cloud storage

Other terms for Big Data Analytics



Source: Figure 3. 2011 TDWI report sponsored by SAS

Why Big Data Analytics

- It improves bottom line
- The economics of analytics is now more embraceable than ever
- It is an enterprise asset that yields actionable business insight
- Analytics based on large data samples reveals and leverages business change

Benefits of Big Data Analytics

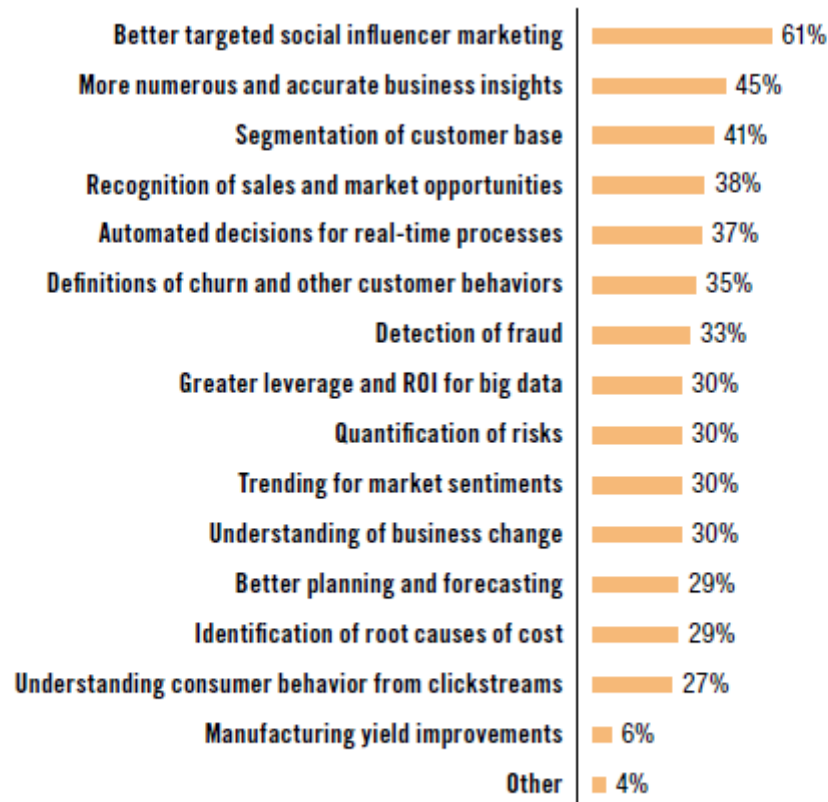


Figure 5. Based on 1,635 responses from 325 respondents; 5 responses per respondent, on average.

Recap

- What is big data?
- What is big data analytics?
- Why should you embrace big data analytics?
- Watch the Intel video (3 minutes)
- <http://bcove.me/vzcvkztu>

Learning Big Data Analytics

- Data Storage, Manipulation, and Retrieval
 - Map Reduce (a parallel computing paradigm)
 - Hadoop (in Java, R, Python, ...)
- Analytics
 - Data on rest (batch computing)
 - Data on stream (using algorithm)
- Data Visualization
 - Communicating your analytic results

MapReduce

- A programming paradigm in which work is broken into mapper and reducer tasks to manipulate data that is stored across a cluster of servers for massive parallelism
- Analogy: how do you find 4 As in a deck of cards? Please use the MapReduce method.

Hadoop

- A top-level Apache project in the Apache Software Foundation written in Java
- A computing environment built on top of a distributed clustered file system that was designed specifically for very large-scale data operation
- Hadoop was inspired by Google (distributed) File system (GFS) and MapReduce programming paradigm

Other Big Data Platforms

- Hadoop (<http://hadoop.apache.org/>)
- BashReduce
(<https://github.com/erikfrey/bashreduce>)
- Disco Project (<http://discoproject.org/>)
- HPCC Systems (<http://hpccsystems.com/>)

(source: <http://www.bytemining.com/2011/08/hadoop-fatigue-alternatives-to-hadoop/>)

- Since this field is fast changing and competition is fierce, an updated list is often required

Data On Rest

- Data rests in a database, data warehouse, a cluster of computing nodes, a cloud
- **Issues of Dark Data**
- Data is available when and if we want it
- Make unstructured data structured
- Chop large problems into small ones
- Mining useful information or association
- Finding significant factors

Data On Stream

- Data arrives so rapidly that it is not feasible to store it all in active storage
- Algorithms should be used to
 - Summarize the stream data
 - Make a useful sample of a stream
 - Filter a stream to eliminate the undesirable elements
 - Use fixed-length window on a stream

Example: Business Big Data Analytics

- Sears used Hadoop cluster to handle data from its Sears, Craftsman, and Lands' End brands to tailor promotions
- “Sears started using the cluster to store incoming data from all its brands and to hold data from existing data warehouses. It then conducted analyses on the cluster directly, avoiding the time-consuming complexities of pulling data from various sources”
- “Sears’s Hadoop cluster stores and processes several petabytes of data at a fraction of the cost of a comparable standard data warehouse”
- (source: Brynjolfsson and McAfee 9-11-2012 HBR)

Example: Process Big Data Analytics

- PASSUR collect flight data every 4.6 seconds for over a decade to predict ETA.
- Minutes matter in airports. So does accurate information about flight arrival times; if a plane lands before the ground staff is ready for it, the passengers and crew are effectively trapped, and if it shows up later than expected, the staff sits idle, driving up costs.
- (source: Brynjolfsson and McAfee 9-11-2012 HBR)

Assignment 1. 10 points

- Read one of the articles in the reading list page. Identify a company that is actively engaging in data analytics. Find more information about this application.
- Write an executive summary for this application.
- A discussion will take place in the Canvas Discussion board. Your extra reading material should be added to the reading list page.